

Will Francis

jf3657@princeton.edu | jwillfrancis107@gmail.com | 858-731-7951 | San Diego, CA

EDUCATION

Princeton University, NJ

Graduation: May 2026

Bachelor of Science in Engineering in Operations Research and Financial Engineering (Applied Math) with minor in Classics

Princeton Varsity Men's Soccer (NCAA Division I): 2024, 2025 Ivy League Tournament Champions; 2x NCAA appearances; recruited as 153rd-ranked player nationally; pre-college Albion SC; Olympic Development Program participant

Relevant Coursework: Probability, Stochastic Systems; Optimization (Foundations & System Applications); Optimal Learning; Networks; Systems Analysis; Computational Models of Cognition; Sports Economics, Multivariable Calculus; Linear Algebra

TECHNICAL SKILLS

Languages & Tools: Python, Java, Stata, Microsoft Excel, SAP, TrackMan, Synergy

DATA ANALYTICS / SPORTS FINANCE EXPERIENCE

R&D Analyst, San Diego Waves, San Diego, CA

Incoming

- Scout opposing and in-house players for behavioral tendencies using TrackMan and proprietary data, translating findings into actionable insights for coaching staff
- Building lineup optimization model from scratch, applying quantitative methods to player sequencing and run-value maximization

Data Analyst, Princeton University Varsity Baseball Team, Princeton, NJ

Jan 2023–May 2026

- Selected through Princeton Baseball Analytics Group to support varsity baseball team with quantitative analysis and scouting
- Conduct pitcher scouting and opponent analysis using high-dimensional pitch-tracking data using velocity, spin, RPM, release, etc.
- Process and clean pitch-level datasets, imputation of missing values and aggregation of metrics to generate reliable player profiles
- Build statistical summaries translating pitch metrics into digestible and custom scouting insights for pre-game preparation
- Convert pitch characteristics into TrackMan simulation parameters, supporting hitter training against created opposing pitcher profiles
- Communicate analytical findings to coaching staff, contributing to scouting and preparation decisions throughout the season

Data Analytics & Finance Intern, L'Oréal Global CPG, Tampa, FL

Jun 2025–Aug 2025

- Identified anomalous spend patterns and trend deviations across multi-million-dollar SG&A budgets using statistical analysis of large, multi-dimensional financial datasets
- Engineered dashboards and data visualizations that translated complex cost-center data into clear, decision-ready insights for senior leadership
- Built forecasting models to project budget trajectories and surface optimization opportunities, with recommendations adopted into strategic planning process

Sports Investment & Wealth Management Intern, AWM Capital, Pasadena, CA

Jun 2024–Aug 2024

- Analyzed 100+ potential private equity, venture capital, and real estate investments to support portfolio construction and capital allocation for high-profile professional athletes
- Built and evaluated cash-flow/risk models reflecting income variability, career length uncertainty, and liquidity needs across portfolios
- Supported tax planning and portfolio rebalancing across asset classes to improve long-term capital efficiency during/after playing careers
- Translated quantitative analysis into practical recommendations used by advisors to manage athlete portfolios under variable performance, markets, and income outcomes

QUANTITATIVE RESEARCH

Swing Time in Division I Baseball: Bridging Bat Speed and Swing Length to Predict Hitter Success

Princeton University Senior Thesis, Department of Operations Research & Financial Engineering, completed April 2026

Advisor: Professor Alain Kornhauser

- Developed "Swing Time" metric via frame-by-frame Synergy video analysis synchronized with TrackMan pitch data, manually collecting 560+ swing observations across 28 Division I hitters spanning multiple conferences
- Demonstrated Swing Time is statistically independent of game context, pitch type, and hitter archetype ($R^2 < 0.05$ across count, score, and base state), validating it as a stable, player-specific mechanical signature
- Applied PCA to MLB Statcast bat-tracking data discovering a 1:1 standardized ratio of bat speed to swing length, with PC2 (42.7% variance) identified as the primary Swing Time dimension
- Built ridge regression and single-hidden-layer neural network to predict season-level AVG, OBP, SLG, OPS, and JOPS from Swing Time distribution features
- Conducted bootstrapped sample size optimization via KS testing, determining that 20 swings per hitter achieves $\geq 90\%$ distributional accuracy, reducing data collection time while preserving model quality

Computational Cognition Modeling Researcher, Princeton University

Depts. of Psychology, Computer Science — Term Research Project

- Modeled hitter swing/take decisions as binary choice under time pressure using deadline-constrained Drift Diffusion Model (DDM)
- Built a pitch-level dataset (~50,000 four-seam fastballs) to isolate perceptual and contextual inputs in rapid decision-making
- Parameterized drift rate as function of pitch characteristics and game context information, enabling decomposition of decision drivers
- Estimated DDM parameters via maximum likelihood, analyzing how different info sources influence evidence toward decisions
- Identified pitch location outside the strike zone ($\beta = -7.50$) as the dominant driver of take decisions, with swing likelihood increasing monotonically across most count states
- Found bases loaded as the only runner state suppressing swing probability, suggesting hitters overweight walk value and revealing a quantifiable, correctable bias in decision-making